Relational Actualism: Irreversible Events as the Primitive Basis of Physical Reality

Relational Actualism Paper V — The Complete Framework

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April 2026

Abstract

Relational Actualism (RA) is a framework in which irreversible on-shell actualization events — physical interactions producing a permanent increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > \Delta S^* = 0.60069\,\text{nats}$ — are the primitive physical reality. Spacetime is the growing directed acyclic graph (DAG) of these events. From a single ontological commitment, without free parameters, RA derives: $\alpha_{\text{EM}} = 1/137$ **[LV]**, $\alpha_s = 1/\sqrt{72}$ **[CV]**, the Koide formula $K = 2/3$ **[LV]**, the complete Standard Model spectrum in five topology classes **[LV]**, $\Lambda = 0$ structurally **[DR]**, the BMV null prediction ($\tau_{\text{act}} \approx 1.9 \times 10^4\,\text{s} \gg \tau_{\text{exp}}$) **[DR]**, the Hubble tension ($H_0^{\text{Eridanus}} \approx 75.9\,\text{km/s/Mpc}$) **[CV]**, the WIMP prohibition **[AR]**, the Boltzmann Brain prohibition **[DR]**, the minimum viable universe seed ($N_{\text{min}} = 2$ connected, $V = +9$) **[DR]**, the universal decoherence timescale ($\tau_d = 20\,\text{fs}$ at $300\,\text{K}$) **[DR]**, the Kinematic Coherence Bound ($N_{\text{max}} = \eta \cdot p_{\text{th}}$) **[LV]**, and the Causal Firewall origin-of-life threshold (same $N_{\text{min}} = 2$ as the Big Bang) **[DR]**. Approximately 110 results are formally verified in Lean 4 with zero sorry tags in the core files. All claims are explicitly marked with epistemic status.

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1 The Single Commitment

1.1 The Ontological Commitment

The Relational Actualism Commitment: An interaction constitutes a *physical event* if and only if it produces an irreversible increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > \Delta S^* = 0.60069$ nats relative to the vacuum state σ_0. Such events write permanent directed edges into the growing causal DAG. Everything else is derived.

1.2 Time and the Arrow

Time is not a background dimension. Time is the process of the causal DAG growing. The arrow of time is not explained by low-entropy initial conditions — it is encoded in the framework's most primitive commitment, since actualization is by definition irreversible. Proper time along a worldline is the integer count of actualization events along that path. Relativistic time dilation is the statement that different worldlines accumulate events at different rates.

1.3 The BDG Action

The dynamics are governed by the Benincasa–Dowker–Glaser action in $d = 4$:

$$S = 1 - N_1 + 9N_2 - 16N_3 + 8N_4 \tag{1}$$

where N_k counts causal-set elements at depth k from candidate vertex v. The integers $(1, -1, 9, -16, 8)$ are uniquely determined by the requirement that this operator converges to the d'Alembertian in the Lorentzian continuum limit [5]. They are *not* free parameters.

2 The Epistemic Map

Every claim in RA carries an explicit epistemic status:

Table 1: Epistemic status categories

Badge	Name	Meaning
[LV]	Lean-Verified	Machine-checked in Lean 4, zero sorry tags. Certainty: as high as mathematics allows.
[CV]	Computation-Verified	Certified numerical computation with explicit error bounds.
[DR]	Derived	All steps explicit in the papers; argument complete but not machine-checked.
[AR]	Argued	Key steps identified; some details incomplete. Upgradeable.
[OP]	Open	Precisely scoped gap with stated path to resolution.

This transparency is intentional. Distinguishing what is *proved* from what is *argued* builds credibility and identifies exactly where theoretical effort should be directed.

3 Lean 4 Formal Verification

3.1 Core Results (Zero Sorry Tags)

The following are machine-verified in RA_D1_Proofs.lean, RA_AmpLocality.lean, RA_Spin2_Macro.l and RA_AQFT_Proofs_v10.lean:

One intentional sorry remains: a Lean-QuantumInfo library adapter in RA_AQFT_Proofs_v10.lean unrelated to any RA claim. Total verified results: ≈ 110.

4 The Coherence Hub

4.1 The Universal Timescale

The universal formula $\tau_{\text{act}} = \Delta S^*/(\Gamma_{\text{on-shell}} \cdot \Delta S_{\text{per boson}})$ is the single expression connecting all scales:

$$T = 1\,\text{mK}: \quad \tau_{\text{act}} \approx 1.9 \times 10^4\,\text{s} \gg \tau_{\text{exp}} \quad \text{(quantum regime — BMV null)} \tag{2}$$

$$T = 300\,\text{K}: \quad \tau_d \approx 20\,\text{fs} = \tau_{\text{vib}} \quad \text{(classical regime — biology)} \tag{3}$$

$$\text{Rindler}: \quad \tau_{\text{act}} \to \infty \quad \text{(KMS: } \Delta S_{\text{per photon}} = 0 \text{ — Unruh resolution)} \tag{4}$$

$$N = 2 \text{ seed}: \quad \tau_d = 0 \quad \text{(instantaneous bootstrap)} \tag{5}$$

$$\text{Spin-bath}: \quad t^* = 0.274/g \quad \text{(parameter-free prediction)} \tag{6}$$

Fourteen orders of magnitude. No free parameters.

Table 2: Lean 4 verified results

ID	Theorem	Statement
L01	LLC	$\Sigma_{\text{out}}(v) = \Sigma_{\text{in}}(v)$ at every DAG vertex
L02	Graph Cut	LLC holds independently on each side of causal severance
L03	Markov Blanket	Subgraph shielding from exterior
L04	Frame Independence	$S(U\rho U^\dagger \| \sigma_0) = S(\rho \| \sigma_0)$ for all Poincaré U
L05	Rindler Stationarity	$\frac{d}{dt} S(\alpha_t(\rho_\beta) \| \sigma_0) = 0$ under modular flow
L06	Rindler Thermal	ρ_β is a valid density matrix
L07	Causal Inv. (cond.)	$\mu_\sigma(S) = \mu_{\sigma'}(S)$ given amplitude locality
L08	α_{EM}	$144 - 7 = 137$ by `norm_num`
L09	Koide	$K = 2/3$ from BDG integers
L10	Confinement	$L_{\text{gluon}} = 3$, $L_{\text{quark}} = 4$
L11	BDG Closure	5 topology types, 124 extensions verified
L12	Qubit Fragility	Electrons/photons at minimum BDG score 1
O01	Amplitude Locality	$a(v\|C)$ depends only on $C \cap \text{past}(v)$
O02	Causal Inv. (full)	Invariant for any permutation $\sigma.\text{Perm}\ \sigma'$
O10s	Spin-2 DOF	$5 - 1 - 1 - 1 = 2$ physical degrees of freedom

4.2 The $N_{\min} = 2$ Identity

The minimum viable universe seed (Paper III) and the Causal Firewall origin-of-life threshold (Paper IV) are the same condition: $N \geq 2$ causally connected actualization events with a directed edge. The Big Bang and the origin of life share the same physical threshold.

4.3 The D4U02 Hub

The D4U02 analytic proof **[DR]** (Paper I, Theorem 1) established that the modular barrier $9 \equiv 1 \pmod 8$ is specific to $d = 4$. This single result unlocks five others: (1) dimensional uniqueness from the conditioning method; (2) O05 (Unruh detector, via τ_{act}); (3) O08 (Firewall τ_d, same formula at molecular scale); (4) new perturbative path for O03; (5) the modular invariants question for Lorentz emergence.

5 Quantum Information

5.1 Kinematic Coherence Bound

Theorem (L12) [LV]: Electrons and photons have minimum BDG stability score 1 (`structural_fragility`, RA_D1_Proofs.lean).

Each qubit is a kinematic site for spurious actualization. The structural ceiling on fault-tolerant quantum arrays:

$$N_{\max} = \eta \cdot p_{\text{th}} \tag{7}$$

where η is the single-qubit quality factor and p_{th} the fault-tolerance threshold. Beyond $N_{\max}$, the cumulative probability of spurious actualization exceeds the fault-tolerance budget regardless of single-qubit fidelity. Standard QM predicts exponentially difficult but unbounded scaling; RA predicts a hard ceiling. Near-future QEC experiments at 10^2–10^3 logical qubits will test this.

6 Open Problems and New Avenues

Table 3: Open problems with proof paths

ID	Problem	New avenue (2026)
O03	Bianchi on curved background	Perturbative $(l_P/R)^2$ expansion; curvature correction bounded by LLC range (L01+L02) — no Tomita-Takesaki needed at leading order
O06	l_{RA} from first principles	Dominant $k = 2$ term in D4U02 corresponds to minimal BDG diamond: area $= l_P^2 \Rightarrow l_{\text{RA}} = l_P$
O10-G	Lean: $G_{\mu\nu}$ derivation	RA-native route: D04 (Bianchi flat) + O10-spin2 (LV) directly, bypassing Weinberg's S-matrix theorem
O11	Lorentz invariance from DAG	BDG modular invariants (c_k mod gcd) as discrete Lorentz-invariants; new question for Dowker collaboration

7 All Predictions

8 The Laws are the Phenomena

Physics has always assumed a hierarchy: laws govern phenomena. RA dissolves this. The Local Ledger Condition is not a constraint that actualization events satisfy; it *is* what actualization events are, stated as a bookkeeping property. Conservation of energy is not imposed from above; it is the LLC read at the appropriate scale. The arrow of time is not explained by initial conditions; it is encoded in the definition of actualization.

Table 4: RA predictions with experimental status

ID	Prediction	Status	Experiment
P01	BMV null; $\tau_{\text{act}} \gg \tau_{\text{exp}}$ at 1 mK	**[DR]**	BMV groups, near-term
P02	$H_0 \propto \rho_b$(LOS); 75.9 at Eridanus	**[CV]**	DESI, current
P03	No DM below neutrino floor	**[AR]**	LZ/XENONnT, ongoing
P04	$t^* = 0.274/g$ (spin-bath)	**[CV]**	SC circuits
P05	$N_{\max} = \eta p_{\text{th}}$ (QEC ceiling)	**[LV]**	QEC, medium-term
P06	Life wherever F1+F2; substrate-independent	**[AR]**	SETI, long-term
P07	$\Sigma m_\nu \approx 59$ meV, normal hierarchy	conj.	CMB-S4/Euclid
P08	No proton decay (baryon number topological)	**[DR]**	—
P09	$\theta_{\text{QCD}} = 0$ exactly (DAG acyclic)	**[DR]**	—

The open problems (Table 3) are not independent mysteries. Each is the same question — "when does a process become irreversible?" — asked at a different scale. The answer is always $\Delta S^* = 0.60069$ nats, because that question has only one answer when irreversibility is the primitive.

Acknowledgments

This programme was developed over twenty-three years and formalized over approximately twelve months with AI collaboration (Claude, Anthropic; Gemini, Google). The Lean 4 infrastructure references the Lean-QuantumInfo library (Meiburg/Perimeter). All papers open-access at https://zenodo.org/communities/relational-actualism.

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